9. Loops : Flow Control

Loops Basics :-

repeat → loops

(a) while loop
(b) for loop
(c) do-while loop
(e) for each loop

While loop :-

Syntax :-
```
while (condition) {
        // Statements
}
```

- Print HW 100 times

```
int counter = 0;
while (counter < 100) {
        Sopln("Hello world");
        counter ++;
}
```

✧ Note : If we write true condition in loops they becomes infinite loops.

Print numbers from 1 to 10 (using while loop):-

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Code :-

```
public class JavaBasics1 {
    public static void main (String [] args) {

        int Counter = 1;
        while (counter <= 10) {

            System.out.print (counter + " ");
            counter ++ ;
        }
        Sopln ();
    }
}
```

→ Output : 1 2 3 4 5 6 7 8 9 10

□ Print number from 1 to n (n = 7)

Code :-

```
import java.util.* ;

public class JavaBasics {
    public static void main (String [] args) {

        Scanner sc = new Scanner (System.in);
        int range = sc.nextInt ();
        int counter = 1;
```

```
        while (counter <= n) {
                System.out.print(counter + " ");
        }
        Sopln();
    }
}
```

→ Output :- n = 7
1 2 3 4 5 6 7

□ Print sum of first n natural numbers :

n = 5 ⇒ 1+2+3+4+5 = 15

```
int n
int sum = 0
int i = 1;
while (i <= n){
        sum = sum+i; // or sum += i
    i++
}
```

Code :-

```
import java.util.*;

public class JavaBasics {
    public static void main (String [] a) {
```

```
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int sum = 0;

        int i = 1;

        while (i<=n){
                sum # = i;
                i++
        }
        System.out.println("Sum is"+ sum);
    }
}
```

→ Output :- 5 (input)
Sum is 15

☐ For loop :-

Syntax :-

```
for (initialization ; condition ; updation){
    for ( i=1 ; i<=10 ; i++)

        //do something

}
```

Print square pattern

```
* * * *
* *  * *
* * * *
* * * *
```

Code :-

```
public class JavaBasics {
    public static void main (String a[]) {

        for (int i=1; i<=4; i++) {

            System.out.println("****");
        }
    }
}
```

Print reverse of a number :-

n = 10889 → 99801

logic :-

① last digit $\xrightarrow{\text{remainder}}$ num % 10

② Remove digit num/10

Code :-

```
public class JavaBasics {
    public static void main (String [] args) {
```

```
int n = 10899;

while (n>0) {
    int lastDigit = n%10

    System.out.print (lastDigit);

    n = n/10;
}
System.out.println();
}
}
```

→ Output:- 99801

□ Reverse the given number

n = 10899 rev = (rev * 10) + lastDigit

Last digit = 9 rev = 0

n = 1089 rev = (0 * 10) + 9
n = 108 rev = (9 * 10) + 9 = 90 + 9 = 99
n = 10 rev = (99 * 10) + 8 = 990 + 8 = 998
n = 1 rev = (998 * 10) + 0 = 9980
rev = (9980 * 10) + 1 = 99801

Code:-

```
public class JavaBasics{
    public static void main (String [] a){

        int n= 10899;
        int rev = 0;

        while (n >0) {

            int lastDigit = n % 10;
            rev = (rev *10) + lastDigit;
            n = n /10;
        }
        Sopln(rev);
    }
}
```

→ Output :- 99801

- Do - while loop :-

Syntax :-

```
do {
    // do something
}
while (condition);
```

- Code :-

```
import java.util.*;

public class JavaBasics{
    public static void main (String [] args){

        int counter = 1;
        do {
            Sopln("Hello world");
            counter++;
        } while (counter <= 5);
    }
}
```

→ Output :- Hello world
Hello world
Hello world
Hello world
Hello world

- Break statement :- To Exit the loop (innermost)

```
for(int i=1; i<=5; i++){

    if (i==3) {
        break;
    }
}
```

Question :- Break Statement :-

Keep entering numbers till user enters a multiple of 10

1	1
5	5
7	7
10	X

Exit

Code :-

```
import java.util.Scanner;
public class JavaBasics{
    public static void main (String [] args){

        Scanner sc = new Scanner(System.in);
        do {
            Sop ("Enter your number:");
            int n = sc.nextInt();

            if (n %10 == 0) {
                break;
            }
            Sopln (n);
        } while (true);
    }
}
```

Output :-

Enter your no = 4
4
Enter your no = 7
7
Enter your no = 50
X Exit loop

Continue Statement :- To skip an iteration

let 1 to 5

1 2 3 4 5

i==3 → continue

Code :-

```
import java.util.*;

public class JavaBasics{
    public static void main (String [] args){

        for(int i=1 ; i<= 5 ; i++){
            if (i==3) {
                continue;
            }
            Sopln (i);
        }
    }
}
```

→ output :- 1
2
4
5

Question : continue Keyword :-

Display all numbers by user except multiple of 10.

Input	Output
1	1
5	5
50	-
7	7
30	-

Code :-

```
import java.util.*;
public class JavaBasics {
    public static void main (String [] args){

        Scanner sc = new Scanner (System.in);

        do {
            Sop ("Enter your number :");
            int n = sc.nextInt ();

            if (n % 10 = =0) {
                continue;
            }

            Sopln("Number was :"+n);
        } while (true);
    }
}
```

Note : for big value :- $\sqrt{n} < (n-2)$

- Check if a number is prime or not

prime : 2, 3, 5, 7, 11

(n) prime = $1 \times n$
$n \times 1$

$n = a \times b$

$12 = 1 \times 12$
2×6
3×4 } unique

4×3
6×2
12×1 } Repeat

$\boxed{n = \sqrt{n} \times \sqrt{n}} \rightarrow 1$ to $\sqrt{n}$

$\boxed{n = i \times b}$

check $\rightarrow$ 2 to $n-1$ [for 12 = 2 to 11]

Code :-

```
import java.util.*;

public class JavaBasics{
    public static void main (String [] args){

        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt ();

        if (n==2){

            Sopln("n is prime");
        }
        else{
            boolean isprime = true;
```

i < Math.Sqrt(n)

```
for(int i = 2; i <= n-1 ; i++) {

        if (n % i == 0) {
            isPrime = false;
        }
    }

    if (isPrime == true){
        Sopln ("n is prime");
    }
    else{
        Sopln("n is not prime");
    }
  }
 }
}
```

→ output :- 7

n is prime

Loops Questions :-

Q.1 How many time "Hello" is print :-

```
public class solution{
    public static void main (String [] args){

        for( int i = 0; i<5 ; i++){

            Sopln ("Hello");
            i += 2;
        }
    }
}
```

Ans 2 times

Q.2 Write a program that reads a set of integer and then prints the sum of even and odd integer.

Sol^n

```
import java.util. Scanner;

public class loops{
    public static void main (String [] a){

        Scanner sc = new Scanner(System.in);

        int number;
```

```
int choice;
int evenSum = 0;
int oddSum = 0;

do{
    Sop ("Enter the number:");
    number = sc.nextInt();

    if (number % 2 == 0){

        evenSum += number;
    } else {
        oddSum += number;
    }

    Sop ("Do you want to continue? Press 1
        for yes or 0 for no");

    Choice = sc.nextInt();
} while (choice == 1);

Sopln("sum of even no" + evenSum );
Sopln("Sum of odd no" + Oddsum );
}
}
```

→ Output:- Enter the number: 4
Do you --- no. 1
Enter the number: 2
Do you want -- no. 1

Enter the number 7
Do you want -- no 0

Sum of even numbers : 6
Sum of odd numbers : 10

Q.3 Write a program to find the factorial of any number entered by the user.

Ans

```
import java.util.Scanner;
public class loops {
    public static void main (String [] args){

        Scanner sc = new Scanner(System.in);

            int num;
            int fact = 1;

        Sop ("Enter any positive integer :");
        num = sc.nextInt();

        for(int i=1; i<=num; i++){

                fact = fact * i;
        }
        Sopln("Factorial :" + fact);
    }
}
```

→ output :- Enter any positive integer : 5
Factorial : 125

Q.4 Write a program to print the multiplication table of a number N, Entered by the user.

Sol^n

```
import java.util.*;
public class loop {
    public static void main (String [] a) {

        Scanner sc = new Scanner (System.in);

        int n;
        Sopln("Enter the number n:");
        n = sc.nextInt();

        for(int i=1; i<=10; i++) {
            Sopln(n+"*"+i+"=@"+(n*i));
        }
    }
}
```

→ Output :- Enter the number n : 5

5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50

Q.5 What is wrong in the following code:-

```
public class loop {
    public static void main (String [] a){
        for (int i=0; i<=5; i++){
            Sopln ("i="+ i);
        }
        Sopln(" i after the loop =" + i);
    }
}
```

→ output:-

Scope of the Variable is refered to the part of the program where the Variable can be used.